Notation

Symbol	Units	Description	Formula
a	1/rad	Lift curve slope ($dC_L/d\alpha$, $dC_L/d\delta$)	
$Æ$	-	Lifting Surface Aspect Ratio	$Æ = b^2/S$
b	m	Lifting surface span	
c	m	Chord length	
c_s	m/s	Speed of sound	
c'/c	-	High-lift device chord extension ratio	
$\overline{c}$	m	Mean Aerodynamic Chord length	$\overline{c} = 2\int_0^{b/2} c^2 dy/S$
C_D	-	Drag Coefficient	$C_D = C_{D_0} + C_{D_i}$
C_{D_0}	-	Drag Coefficient at zero lift	
C_{D_i}	-	Lift dependent/induced Drag Coefficient	$C_{D_i} = {C_L}^2/\pi Æe$
C_H	-	Hinge Moment Coefficient	
C_L	-	Lift Coefficient	
C_{L_0}	-	Lift Coefficient at zero aircraft angle of attack	
C_M	-	Pitching Moment Coefficient	
C_{M_0}	-	Pitching Moment Coefficient at zero lift (constant when taken about surface aerodynamic centre)	
C_T	-	Thrust Coefficient	$C_T = T/0.5\rho {V_\infty}^2 S$
D	N	Drag Force	$D = 0.5\rho {V_\infty}^2 S C_D$
e	-	Oswald/span efficiency factor	
g	m/s^2	Gravitational acceleration	$g = 9.81$ m/s^2
H	N m	Hinge moment	$H = 0.5\rho {V_\infty}^2 S_{surf}\overline{c}_{surf} C_H$
H_m	-	Stick-fixed manoeuvre margin	
I_{xx}	kg m^2	Roll second moment of inertia	$I_{xx} = \int (y^2 + z^2) dm$
I_{yy}	kg m^2	Pitch second moment of inertia	$I_{yy} = \int (x^2 + z^2) dm$
I_{zz}	kg m^2	Yaw second moment of inertia	$I_{zz} = \int (x^2 + y^2) dm$
I_{xz}	kg m^2	Cross second moment of inertia	$I_{xz} = \int xz dm$
i	rad	Lifting surface setting angle - defined between mean aerodynamic chord and aircraft z-datum	
K_n	-	Stick-fixed static margin	$\overline{x}_{np} - \overline{x}_{CG}$
K'_n	-	Stick-free static margin	$\overline{x}'_{np} - \overline{x}_{CG}$
L	N	Lift	$L = 0.5\rho {V_\infty}^2 S C_L$
L	N m	Rolling Moment (+ve port wing up)	
l_H	m	Horizontal separation between wing and tailplane aerodynamic centres	$x_H - x_W$
l'_H	m	Horizontal tailplane lift moment arm	$x_H - x_{CG}$
m	kg	Aircraft mass	
M	N m	Pitching Moment (+ve nose up)	$M = 0.5\rho {V_\infty}^2 S\overline{c} C_M$
M_∞	-	Freestream Mach number	$\mathrm{M}_\infty = V_\infty/c_s$
n	-	Load factor (n = Lift/Weight)	
N	N m	Yawing moment (+ve port wing forward)	

p	rad/s	Roll rate (+ve port wing up)	
q	rad/s	Pitch rate (+ve nose up)	
q_∞	Pa	Dynamic head/pressure	$q_\infty = 0.5\rho {V_\infty}^2$
r	rad/s	Yaw rate (+ve port wing forward)	
S	m^2	Lifting Surface Planform Area	
T	N	Thrust	
T_d	s	Damped Period of Oscillation	$T_d = 2\pi/\omega_d$
T_n	s	Undamped Period of Oscillation	$T_n = 2\pi/\omega_n$
U	m/s	Velocity along x wind axis (+ve moving forward)	
V	m/s	Velocity along y wind axis (+ve moving starboard)	
V_∞	m/s	Freestream Velocity	$V_\infty = \sqrt{U^2 + V^2 + W^2}$
$\overline{V}_H$	-	Horizontal tailplane volume coefficient	$(\overline{x}_H - \overline{x}_W)S_H/S_{ref}$
$\overline{V}'_H$	-	Modified Horizontal tailplane volume coefficient	$(\overline{x}_H - \overline{x}_{CG})S_H/S_{ref}$
$\overline{V}_V$	-	Vertical tailplane volume coefficient	$(\overline{x}_V - \overline{x}_{CG})S_V/S_{ref}$
W	m/s	Velocity along z wind axis (+ve moving downward)	
W	N	Aircraft weight	
x	m	Distance aft of aircraft datum, parallel to freestream velocity vector	
x_{ac}	m	Chordwise position about which surface $dC_M/dC_L = 0$	
x_{np}	m	Distance of stick-fixed neutral point aft of aircraft datum, parallel to freestream velocity vector	
x'_{np}	m	Distance of stick-free neutral point aft of aircraft datum, parallel to freestream velocity vector	
x_W	m	Distance of wing aerodynamic centre aft of aircraft datum, parallel to freestream velocity vector	
X	N	Force along x wind axis (+ve forward)	
y	m	Spanwise distance port of aircraft datum, normal to the $x - z$ plane	
Y	N	Force along y wind axis (+ve to starboard)	
z	m	Distance above aircraft datum, normal to freestream velocity vector	
Z	N	Force along z wind axis (+ve downwards)	

Greek Letters

Symbol	**Units**	**Description**	**Formula**
α	rad	Angle of Attack, defined between freestream velocity and aircraft z-datum	
α_0	rad	Lifting surface zero-lift angle of attack	

β	rad	Sideslip Angle, defined between freestream velocity and aircraft centreline	
Γ	deg	Dihedral angle	
δ	rad	Control surface deflection	
ε	rad	Downwash/upwash angle	
ε_0	rad	Downwash/upwash angle at zero aircraft angle of attack	
ϵ	rad	Thrust setting angle - defined between thrust line and aircraft z-datum	
ζ	-	Damping ratio	
η	-	Lifting surface aerodynamic efficiency factor	$\eta = (V_{local}/V_\infty)^2$
Θ	rad	Pitch angle - defined between wind body-fixed axes and the earth frame (+ve in a climb)	
λ	-	Eigenvalue	
Λ	rad	Lifting surface sweep angle (typically at quarter-chord line)	
ρ	kg/m^3	Density of air	
Φ	rad	Roll angle - defined between wind body-fixed axes and the earth frame (+ve in right roll)	
Ψ	rad	Yaw angle- defined between wind body-fixed axes and the earth frame (+ve in right turn)	
ω	rad/s	Angular velocity	
ω_d	rad/s	Damped natural angular frequency of oscillation	
ω_n	rad/s	Undamped natural angular frequency of oscillation	$\omega_d = \omega_n\sqrt{(}1-\zeta^2)$

Subscripts

Symbol	**Description**
$(\cdot)_A$	Property relating to aileron
$(\cdot)_C$	Property relating to canard
$(\cdot)_{CG}$	Centre of Gravity
$(\cdot)_e$	Property at equilibrium value
$(\cdot)_E$	Property relating to elevator
$(\cdot)_H$	Property relating to horizontal tailplane
$(\cdot)_R$	Property relating to rudder
$(\cdot)_{ref}$	Aircraft reference value (typically $S_{ref} = S_W$, $\bar{c}_{ref} = \bar{c}_W$)
$(\cdot)_T$	Property relating to trim-tab
$(\cdot)_V$	Property relating to vertical tailplane
$(\cdot)_W$	Property relating to main lifting surface (wing)